

Research Article

EMOTION AND INTUITION: Effects of Positive and Negative Mood on Implicit Judgments of Semantic Coherence

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Abstract—We investigated effects of emotional states on the ability to make intuitive judgments about the semantic coherence of word triads. Participants were presented word triads, consisting of three clue words that either were weakly associated with a common fourth concept (coherent triads) or had no common associate (incoherent triads). In Experiment 1, participants in a neutral mood discriminated coherent and incoherent triads reliably better than chance level even if they did not consciously retrieve the solution word. In Experiment 2, the induction of a positive mood reliably improved intuitive coherence judgments, whereas participants in a negative mood performed at chance level. We conclude that positive mood potentiates spread of activation to weak or remote associates in memory, thereby improving intuitive coherence judgments. By contrast, negative mood appears to restrict spread of activation to close associates and dominant word meanings, thus impairing intuitive coherence judgments.

It is a common belief that the way people perceive the world, process information, and solve problems is influenced by emotions and moods. Probably all of us have experienced that when we are in a depressed mood, our attention appears to be focused in a rigid and narrow way, whereas when we are in a happy mood, our minds seem to bubble over with ideas and sometimes far-fetched associations. Such everyday impressions are supported by a growing body of empirical evidence showing that positive and negative moods are in fact accompanied by qualitatively different information processing modes (e.g., Gray, 2001; Isen, 1999; Kuhl, 1983, 2000). For instance, individuals in a positive mood, compared with those in a negative mood, produce more unusual associations (Isen, Johnson, Mertz, & Robinson, 1985), show improved performance on tests of creative problem solving (Isen, Daubman, & Nowicki, 1987), prefer heuristic over exhaustive decision-making strategies (Isen & Means, 1983), and form more inclusive categories in a sorting task (Isen & Daubman, 1984; see Isen, 1999, for review).

In this study, we investigated a hypothesis derived from a more general theoretical framework termed *personality systems interaction theory* (Kuhl, 2000; Kuhl & Kazén, 1999). According to this theory, an increase in positive affect supports a *holistic* processing mode, which is characterized in memory by the activation of wide semantic fields, which include weak or remote associates. In contrast, an increase in negative affect supports an *analytic* processing mode, which is characterized by a more restricted spread of activation to close associates and dominant word meanings. We focus on a specific prediction of this hypothesis—that a positive mood, because of increased activa-

tion of remote associates, improves the ability to make intuitive judgments about the semantic coherence of verbal stimuli, whereas a negative mood impairs intuitive coherence judgments.

We conceive of intuition as the ability to make above-chance judgments about properties of a stimulus on the basis of information that is not consciously retrieved (cf. Bastick, 1982; Westcott, 1968). An example would be the ability to discriminate between pieces of music from the classical and romantic periods, even if one is unable to describe the basis of this judgment verbally. Although the concept of intuition has often been considered elusive, there is now accumulating evidence that various types of judgments are influenced by information that is not accessible to consciousness (for reviews, see Goschke, 1997; Greenwald, 1992; Kihlstrom, 1987; Lieberman, 2000; Perrig & Wippich, 1995; Reber, 1993; Schacter, 1987). Directly related to our investigation is a study of Bowers, Regehr, Balthazard, and Parker (1990), who showed that participants were able to decide at an above-chance level whether or not word triads (e.g., “playing, credit, report”) were coherent (i.e., had a common weak associate—“card”) even if they did not retrieve this solution word consciously. This finding suggests that activation spreads from the memory representation of the clue words in a coherent triad to the solution concept. Even if the solution word does not gain access to consciousness, its subthreshold activation presumably leads to an intuitive perception of coherence that guides participants’ judgments (cf. Bowden & Beeman, 1998; Mednick, 1962; Metcalfe & Wiebe, 1987; Yaniv & Meyer, 1987).

The main purpose of our study was to investigate effects of positive and negative mood on intuitive coherence judgments. From personality systems interaction theory (Kuhl, 2000), we derived the prediction that an increase in positive affect, by potentiating spread of activation to weak associates, should improve the ability to make intuitive coherence judgments, whereas a negative mood, by restricting spread of activation to close associates and dominant word meanings, should impair intuitive coherence judgments. Note that this *affect-modulation hypothesis* is not concerned with the effects of mood on the activation of mood-congruent memory contents (cf. Bower, 1981; Goschke, 1996). Rather, it concerns content-unspecific effects of affective states on qualitatively different (analytic vs. holistic) processing modes (Kuhl, 1983, 2000; cf. Forgas, 2000; Gray, 2001). To test this hypothesis, we had participants perform a modified version of the coherence judgment task with emotionally neutral material either in a neutral mood (Experiment 1) or after the induction of a positive or negative mood (Experiment 2).

EXPERIMENT 1

In Experiment 1, we investigated whether participants in a neutral mood are able to judge the semantic coherence of word triads above

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chance in the absence of conscious knowledge of the solution word. We changed several features of the coherence task used in the original study by Bowers et al. (1990), in order to avoid some methodological shortcomings. Moreover, we used a more conservative criterion for classifying coherence judgments as implicit.

Method

Participants

Participants were 24 undergraduates from the University of Osnabrück, Germany (9 men and 15 women; mean age = 25, range: 22–34).

Materials and apparatus

Stimuli were German translations of 72 word triads taken from the study by Bowers et al. (1990); they had been adapted from the Remote Associates Test (Mednick & Mednick, 1967). Half of the triads were coherent; that is, each of the three clue words was a low associate of a common fourth word (the solution word). The other half of the triads were incoherent; that is, the three words were not associated with a common solution word. Coherent and incoherent triads were matched with respect to the semantic relatedness among the three clue words (cf. Bowers et al., 1990, p. 76). Each participant was presented 12 coherent and 12 incoherent triads.

Procedure

Participants were instructed that they would have to judge whether or not word triads were semantically coherent. It was explained that the clue words of coherent triads were all weakly associated with a common target word. On each trial, the three clue words were presented one above the other at the center of the computer monitor for 4 s. After the offset of the triad, participants indicated their coherence judgment by pressing the “-” or “y” key of the computer keyboard. After each response, participants indicated their confidence in the coherence judgment on a 4-point rating scale (1 = *low confidence*; 4 = *high confidence*).¹ Finally, participants were prompted to enter a solution word via the computer keyboard. When participants were insecure or thought that there was no solution word, they were forced to guess and type in any word that came to mind.

Results and Discussion

Solved triads

We classified a coherent triad as solved if the participant produced the preordained solution word or an exact synonym of the solution word. In addition, we considered a word to be a solution if it was judged to be semantically related to the three clue words by at least one of two independent raters.² We followed this procedure to detect trials on which participants came up with plausible, but unanticipated

solution words (e.g., when “hike,” instead of the preordained solution, “mountain,” was given as a solution for the triad “goat, pass, green”). Across participants, a total of 101 out of 288 coherent triads (35%) were classified as solved; that is, participants solved on average 4.2 of the 12 coherent triads. Of the 101 produced solution words, 37 were preordained solutions, whereas 64 were synonyms of the preordained solutions or words that were semantically related to the clue words. As expected, none of the produced words for incoherent triads was classified as semantically related to the clue words.

Coherence judgments

Solved coherent triads were almost always correctly classified by participants as coherent (see Table 1). For the unsolved triads, we computed *hit rates* (i.e., the proportion of unsolved coherent triads that were correctly classified as coherent) and *false alarm rates* (i.e., the proportion of unsolved incoherent triads that were falsely classified as coherent). Participants correctly classified 61% of the unsolved coherent triads as coherent; this hit rate was significantly above the chance value of 50%, $t(23) = 2.14$, $p < .05$. Participants falsely classified 48% of the incoherent triads as coherent; this false alarm rate did not significantly depart from chance level, $t(23) = -0.50$. As an index for the ability to discriminate between unsolved coherent and incoherent triads, we computed the difference between the hit and false alarm rates. This *intuition index* was 13% and departed significantly from 0, $t(23) = 2.21$, $p < .04$.

Response times

For unsolved triads, correct coherence judgments were made faster than incorrect judgments, both for coherent triads (3,343 vs. 3,597 ms) and for incoherent triads (2,598 vs. 3,268 ms). However, a 2×2 analysis of variance (ANOVA) with triad type (coherent vs. incoherent) and judgment (coherent vs. incoherent) as independent variables yielded no reliable effects.

Conclusions

Participants' performance in judging the semantic coherence of word triads was reliably better than chance level even when they did not consciously retrieve the solution words. We obtained this finding even though we took several precautions to be as conservative as possible in classifying triads as unsolved. First, in addition to preordained

Table 1. Mean number of solved and unsolved coherent and incoherent triads that were classified as coherent or incoherent, Experiment 1

Judgment	Coherent triads		Incoherent triads	
	Solved	Unsolved	Solved	Unsolved
Coherent	3.63 (1.97)	4.42 (1.91)	0.00	5.38 (2.22)
Incoherent	0.58 (0.83)	3.00 (2.00)	0.00	5.88 (2.35)

Note. Standard deviations are given in parentheses. Because of occasional missing values, the sum of the individual frequencies is slightly smaller than the expected value of 24. Incoherent triads by definition had no solutions.

1. In both experiments, average confidence for unsolved triads was relatively high (2.8 and 3.0 on a 4-point scale, respectively) and differed only minimally between coherent and incoherent triads and between mood conditions.

2. The two raters were the first author and a second rater who was blind to whether or not triads were coherent and to whether or not participants had judged a triad as coherent.

solution words and exact synonyms, words that were semantically related to the clue words were considered plausible solutions. Second, because we forced participants to guess when they felt unable to provide a solution word, we can exclude the possibility that participants refrained from reporting candidate solutions because of a strict response criterion. Third, because participants entered the solution word only after making a coherence judgment, we can be sure that participants who did not enter a correct solution were also not aware of a solution when they pondered the coherence judgment. By contrast, in the study by Bowers et al. (1990), participants first searched for a solution and then made their coherence judgment, which leaves open the possibility that participants who initially did not come up with a solution retrieved the solution while pondering the coherence judgment.

In conclusion, our findings provide strong evidence that participants were able to judge semantic coherence on an intuitive basis. One possible explanation of this finding is that although each of the three words of a triad activated various semantically associated concepts, only in the case of coherent triads did activation converge on a common concept (cf. Mednick, 1962). Even if this activation was not sufficient for conscious retrieval of the solution word, it obviously elicited an intuitive impression of semantic coherence, which guided participants' judgments.

EXPERIMENT 2

In Experiment 2, we investigated effects of an induced positive or negative mood on intuitive coherence judgments.

Method

Participants

Participants were 48 undergraduates from the University of Osnabrück (22 males and 26 females; mean age = 25, range: 19–43). Half of the participants were assigned at random to the positive-mood group, and half were assigned to the negative-mood group.

Procedure

At the beginning of the session, the participants' baseline mood was assessed with an adjective checklist comprising separate scales for subjective happiness, sadness, and arousal. Participants used a 4-point scale to indicate whether each adjective described their current state. After a brief relaxation phase, participants in the negative-mood group were instructed to recall as clearly and vividly as possible a sad episode from their personal past, and participants in the positive-mood group were instructed to recall a happy episode. They were encouraged to form a mental image of the episode and to evoke the emotion they had felt during the original episode. Participants were given 5 min for imaging the episode, and then their mood was again assessed with the adjective checklist. Next, they performed the coherence task, with the same materials and procedures as in Experiment 1.

Results and Discussion

Mood induction

As can be seen in Table 2, the positive- and negative-mood groups did not differ with respect to their baseline happiness or sadness (both p s > .30). After the positive-mood induction, participants' mean happiness score reliably increased compared with the baseline score,

Table 2. Mean ratings (%) for the happiness and sadness scales before and after the mood induction, Experiment 2

Measure	Mood group	
	Positive	Negative
Positive mood		
Baseline	51.16 (15.37)	47.45 (25.06)
Postinduction	65.51 (20.66)	24.54 (20.39)
Negative mood		
Baseline	10.59 (15.00)	7.29 (7.08)
Postinduction	3.65 (6.98)	16.32 (15.59)

Note. Standard deviations are given in parentheses. Means were computed for each participant by dividing the summary score on each of the two rating scales by the maximal possible value and multiplying the result by 100.

$t(23) = -3.79$, $p < .01$, whereas their mean sadness score decreased reliably, $t(23) = 3.06$, $p < .01$. After the negative-mood induction, the mean sadness score increased reliably compared with the baseline score, $t(23) = -2.85$, $p < .01$, whereas the happiness score reliably decreased compared with the baseline score, $t(23) = 4.98$, $p < .001$. After the mood induction, participants in the positive-mood group were reliably more happy and less sad than participants in the sad group, $t(46) = 6.92$, $p < .001$, and $t(46) = -3.64$, $p < .01$, respectively. Thus, the mood induction was clearly effective.

Solved triads

As in Experiment 1, triads were considered as solved if participants entered the preordained solution word, an exact synonym, or a word that was judged to be semantically related to the clue words by at least one of two independent raters.³ On average, participants produced a correct solution for 23% (2.8 out of 12) of the coherent triads. Of the 134 solution words that were classified as correct solutions, 63 consisted of preordained solution words, whereas the remaining 71 solution words were either synonyms of the preordained solution word or semantically related to the clue words.

Coherence judgments and mood effects

Solved triads were almost always classified as coherent (see Table 3). For unsolved triads, hit and false alarm rates and the intuition index were computed as in Experiment 1.

A 2×2 ANOVA with mood (positive vs. negative) and triad type (coherent vs. incoherent) as independent variables and the proportion of unsolved triads that were classified as coherent as the dependent variable yielded a highly reliable effect of triad type, $F(1, 46) = 35.53$, $p < .001$, $MSE = 0.02$, as well as a highly reliable interaction of mood and triad type, $F(1, 46) = 23.46$, $p < .001$, $MSE = 0.02$. The main effect of mood was not significant, $F < 1$.

3. Produced words were classified by the first author and a second independent rater. Interrater agreement was almost perfect (98%).

Table 3. Mean number of solved and unsolved coherent and incoherent triads that were classified as coherent or incoherent, Experiment 2

Mood group and judgment	Coherent triads		Incoherent triads	
	Solved	Unsolved	Solved	Unsolved
Positive mood				
Judged as coherent	2.92 (1.74)	5.33 (1.69)	0.00	3.75 (2.01)
Judged as incoherent	0.17 (0.38)	3.38 (1.86)	0.00	8.17 (1.97)
Negative mood				
Judged as coherent	1.96 (1.23)	4.96 (2.07)	0.00	5.50 (2.25)
Judged as incoherent	0.54 (0.88)	4.29 (1.73)	0.00	5.63 (2.36)

Note. Standard deviations are given in parentheses. Because of occasional missing values, the sum of the individual frequencies within each mood group is slightly smaller than the expected value of 24.

In the positive-mood group, the hit rate (i.e., the proportion of unsolved coherent triads that were correctly classified as coherent) was 62%, which was reliably above chance level, $t(23) = 2.94, p < .01$. Moreover, the false alarm rate (i.e., the proportion of unsolved incoherent triads that were incorrectly classified as coherent) was 31% and reliably below chance level, $t(23) = -5.48, p < .001$. The intuition index (i.e., the difference between hit and false alarm rates) was highly reliable, $t(23) = 7.25, p < .001$. In marked contrast, participants in the negative-mood group produced a hit rate of only 53% and a false alarm rate of 50%, both of which did not reliably differ from chance level, $t(23) = 0.68, p = .50$, and $t(23) = -0.08, p = .94$, respectively. Accordingly, the intuition index did not differ reliably from chance, $t(23) = 0.84, p = .41$. A direct comparison between the two mood groups showed that the intuition index was reliably higher for the positive- than for the negative-mood group (31% vs. 3%, respectively), $t(46) = 4.84, p < .001$. Moreover, we computed for each subject a nonparametric signal detection measure of discriminability, termed A' (Pollack, 1970).⁴ In the positive-mood group, the mean value of A' was .56, which was reliably above chance level (.50), $t(23) = 5.27, p < .001$; in contrast, A' was at chance level (.50) in the negative-mood group. The value of A' was reliably higher in the positive- than in the negative-mood group, $t(46) = 4.33, p < .001$.

Mood effects were also reflected in a positive correlation between self-rated happiness and the intuition index, $r = .39, p < .01$. Self-rated arousal was not correlated with the intuition index ($r = -.09, p > .50$). An analysis of covariance with the intuition index as the dependent variable, mood induction as the independent variable, and arousal as a covariate yielded a highly reliable effect of mood induction, $F(1, 45) = 20.35, p < .00, MSE = 0.039$, but no effect of the covariate ($F < 1$). It is thus unlikely that mood effects were mediated by changes in arousal.

4. If H is the hit rate and FA the false alarm rate, then A' is calculated according to the following formulas:

$$\text{If } H > FA, A' = .5 + (H - FA) / (1 + H - FA) / 4H(1 - FA)$$

$$\text{If } H = FA, A' = .5$$

$$\text{If } H < FA, A' = .5 - (FA - H) / (1 + FA - H) / 4FA(1 - H).$$

Chance performance yields an A' of .5; perfect performance yields an A' of 1.0.

Comparison of Experiments 1 and 2

As the instructions, materials, and procedures of Experiments 1 and 2 were completely identical apart from the mood induction, it is informative to compare the intuition indices across the two experiments (see Fig. 1). A one-way ANOVA yielded a highly reliable main effect of mood group (neutral vs. negative vs. positive), $F(2, 69) = 8.56, p < .001, MSE = 0.05$. Scheffé tests showed that the intuition index was reliably higher in the positive-mood group than in the neutral group (31% vs. 13%, $p < .01$), whereas it was reliably smaller in the negative-mood compared with the neutral-mood group (3% vs. 13%, $p < .001$). Coherence judgments were thus improved in a positive mood as well as impaired in a negative mood.

Response times

Correct judgments were made faster than incorrect judgments, both for coherent triads (2,471 vs. 3,261 ms) and for incoherent triads (2,697 vs. 3,034 ms). In an ANOVA with mood induction, triad type (coherent vs. incoherent), and judgment (correct vs. incorrect) as independent variables and mean response time for unsolved triads as the dependent variable, the only reliable result was a main effect of judgment, $F(1, 43) = 7.58, p < .01, MSE = 2,313,522$.

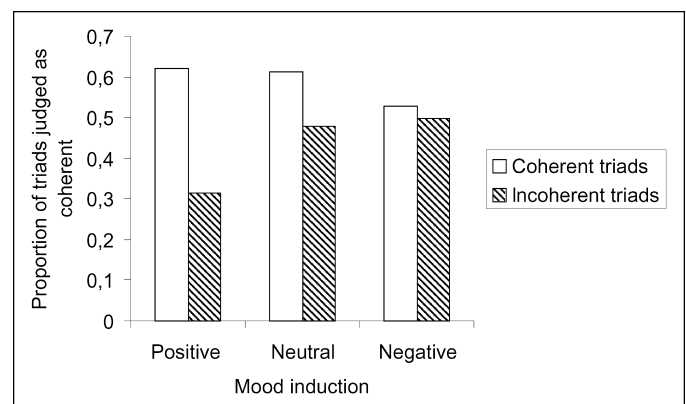


Fig. 1. Proportion of unsolved coherent and incoherent triads that were classified as coherent by participants in a neutral, positive, or negative mood (data from Experiments 1 and 2).

Conclusion

We obtained a strong effect of mood on the ability to make intuitive coherence judgments. Participants in a positive mood discriminated unsolved coherent and incoherent triads reliably better than participants in a negative or neutral mood. Strikingly, participants in a negative mood performed at chance level.

GENERAL DISCUSSION

In two experiments, participants performed reliably better than chance in discriminating word triads in which the three clue words either did or did not have a common associate, even if they did not consciously retrieve the solution word. One possible interpretation of this finding is that in the case of coherent triads, activation spread from the three clue words to the common associate and that this subthreshold summation of activation gave rise to an intuitive impression of semantic coherence (Bowers et al., 1990; Dorfman, Shames, & Kihlstrom, 1996; Mednick, 1962). Note that participants were faster when they classified unsolved coherent triads correctly than when they falsely classified such triads as incoherent. It is therefore unlikely that incorrect responses reflected overly hasty judgments, whereas correct judgments involved a more extended problem-solving process. To the contrary, correct judgments were obviously guided by a relatively rapidly evolving intuitive impression of coherence. This conclusion is supported by a recent study in which we found that participants made above-chance intuitive coherence judgments even if they were forced to respond within 1.5 s after the appearance of a triad (Bolte, 1999).

Most important, induced mood had a strong effect on intuitive coherence judgments. A positive mood reliably improved participants' ability to make intuitive coherence judgments, whereas participants in a negative mood performed at chance level. This result is consistent with our affect-modulation hypothesis (Kuhl, 2000), according to which positive mood promotes the activation of widespread associative networks, whereas negative mood restricts the spread of activation, such that remote associates are not sufficiently activated to guide intuitive coherence judgments (cf. Fiedler, 1988; Isen, 1999). In a study consistent with this interpretation, Baumann and Kuhl (2002) found that when participants were in a sad mood, those with a reduced ability to cope with negative affect (so-called state-oriented participants) performed worse on the coherence task than those who tended to down-regulate negative emotions (so-called action-oriented participants).

Although our results are consistent with the assumption that mood modulates the degree to which activation spreads to remote associates, there are alternative interpretations that demand consideration. Perhaps positive mood increases participants' optimism and confidence, thus making them more willing to base coherence judgments even on a vague feeling of coherence. By contrast, negative mood may increase pessimism and reduce confidence, making participants more cautious and less likely to rely on intuitive impressions. This assumption would account for improved coherence judgments among participants in a positive mood even if there was no differential activation of solution concepts in positive and negative moods. However, contrary to this interpretation, happy and sad participants showed virtually identical confidence in their coherence judgments for correctly classified unsolved triads (3.17 and 3.14, respectively). Moreover, a signal detection analysis clearly showed that mood affected discriminability independently from changes in response criteria.

Another interpretation is suggested by the assumption that mood serves an informational function in that it indicates whether a situation is benign or problematic, thereby tuning cognitive strategies to meet the appropriate situational requirements (e.g., Schwarz, 2001). We do not consider this assumption to be incompatible with our spreading-activation account. If a positive mood signals the absence of problems or obstacles, it may inform the organism that there is no risk in engaging in an intuitive, holistic, or creative mode of thought. By contrast, a negative mood usually indicates the presence of conflicts, problems, or dangers, and may thus signal that a more analytic problem-solving mode of processing is required. Thus, mood effects on the activation of remote associates may well reflect the informational function of mood states. We assume, however, not only that the informational content of mood influences cognitive processes by way of a deliberate strategy change, but also that mood can modulate cognitive processing modes in a more automatic way, as indicated by our finding that mood influenced intuitive coherence judgments independently from changes in response criteria.

An important question for future research concerns the neural systems underlying modulatory effects of affective states. At present, we do not know whether activation of close associates and spread of activation to remote associates reflect different processes within a single representation system, or whether they are mediated by separate systems. Suggestive evidence for the latter possibility stems from studies indicating that the two cerebral hemispheres subserve different forms of semantic representations. Notably, Beeman et al. (1994) found larger semantic priming effects for target words that were preceded by three weakly related primes when the targets were presented in the left visual hemifield (presumably increasing initial right-hemispheric processing) than when they were in the right visual field (see also Abdullaev & Posner, 1997; Chiarello & Richards, 1992; Kiefer, Weisbrod, Kern, Maier, & Spitzer, 1998). Moreover, a recent neuroimaging study (Seeger, Desmond, Glover, & Gabrieli, 2000) showed increased activation in left inferior prefrontal cortex when participants generated usual (typical) verbs in response to novel nouns, whereas increased activation was observed in right middle and superior frontal gyri when participants generated unusual (atypical) verbs. The authors concluded that the left hemisphere is biased toward processing close associates, whereas the right hemisphere may be involved in the activation of distant associates and creative problem solving. Bowden and Beeman (1998) found that participants who had unsuccessfully tried to solve insight problems showed greater priming for solution-related information presented to the left visual field than for information presented to the right visual field. In light of these observations, an intriguing hypothesis for future research is that the mood effects we obtained may reflect a change in the relative balance of left- versus right-hemispheric involvement in semantic processing (cf. Kuhl, 2000; Tucker, 1981).

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